

**FMUSER FUQ-3224K/FUQ-3224GJ 4-8-12 Channel HD-SDI-HDMI  
1.4 Encoder HEVC-H.265 H.264 Gigabit IP Output Web-Based  
Control for Broadcast IPTV**



- ◆ **Video Encoding:** HEVC/H.265 & MPEG 4 AVC/H.264
- ◆ **Audio Encoding:** MPEG-1 Layer 2, LC-AAC, HE-AAC, HE-AAC V2, AC3 Passthrough
- ◆ **Ultra Low Bit Rate:** Save 75% Bandwidth
- ◆ **Enhance Picture Quality:** Advanced Compressing Algorithm
- ◆ **Advanced Pretreatment:** De-interlacing, Noise Reduction, Sharpening
- ◆ **OSD (Logo, QR Code) insertion**—Optional as per order



**News Channel/Movies**

**1Mbps Full HD**



**B frame(IBBP) GOP Structure**



**Sports Channel**

**2Mbps Full HD**



**HDMI 1.4**



**Full HD 1080P**



**HDCP 1.4**



**Decoder/STB**

**STB Available with Ensurity CAS**

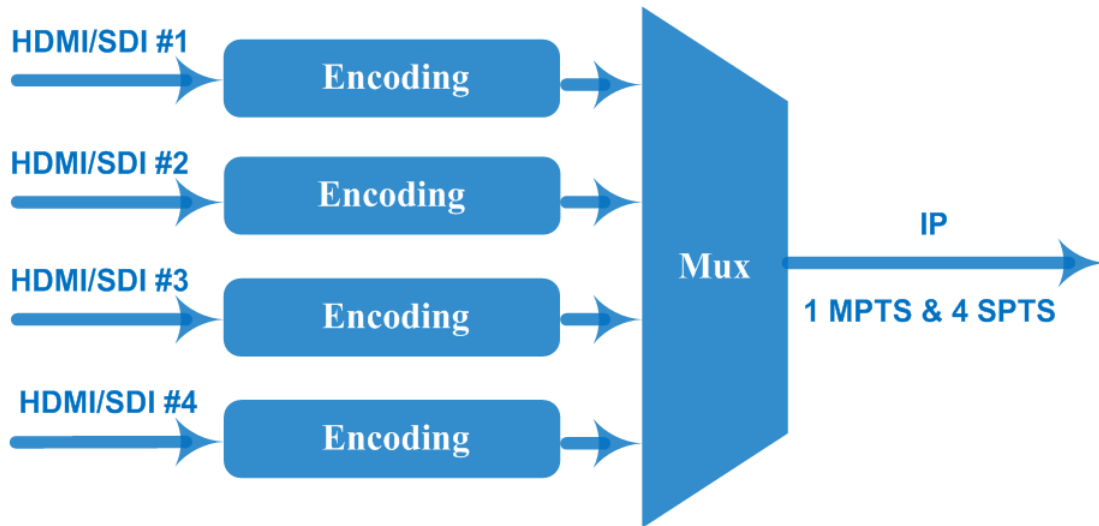
**Decoding Chipset: Montage CS8051**

**NationalChip GX3201H**

## OSD (Logo/PPT/Caption/QR Code) Insertion



Principle Chart of Per Module



### Product overview

The FUQ-3224K/FUQ-3224GJ modular encoder processes 4/8/12×HD-SDI/HDMI 1.4 inputs with HDCP 1.4 compliance. It encodes signals to HEVC/H.265 or MPEG-4 AVC/H.264 formats at 0.5-20Mbps, outputting via Gigabit IP (UDP/RTP/RTSP). The 1RU chassis supports 4K-ready processing and web-based control. Ideal for live event production, IPTV headends, and remote contribution systems.

### Key feature

Web-based management with Chinese/English UI and 48kHz audio processing

### Technical Specification

| Input             | 4/8/12×HD-S<br>DI or<br>HDMI(1.4)<br>input for<br>option, HDCP<br>1.4 | 4/8/12×HD-S<br>DI or<br>HDMI(1.4)<br>input for<br>option, HDCP<br>1.4 | 4/8/12×HD-S<br>DI or<br>HDMI(1.4)<br>input for<br>option, HDCP<br>1.4 | 4/8/12×HD-S<br>DI or<br>HDMI(1.4)<br>input for<br>option, HDCP<br>1.4 | 4/8/12×HD-S<br>DI or<br>HDMI(1.4)<br>input for<br>option, HDCP<br>1.4 |
|-------------------|-----------------------------------------------------------------------|-----------------------------------------------------------------------|-----------------------------------------------------------------------|-----------------------------------------------------------------------|-----------------------------------------------------------------------|
| Video<br>Encoding | Encoding<br>Format                                                    | HEVC/ H.265,<br>MPEG 4<br>AVC/H.264                                   | HEVC/ H.265,<br>MPEG 4<br>AVC/H.264                                   | HEVC/ H.265,<br>MPEG 4<br>AVC/H.264                                   | HEVC/ H.265,<br>MPEG 4<br>AVC/H.264                                   |
| Video<br>Encoding | Resolution<br>(DX3224HV)                                              | Input                                                                 | Output                                                                | Output                                                                | Output                                                                |
| Video<br>Encoding | Resolution<br>(DX3224HV)                                              | Input                                                                 | HEVC/H.265                                                            | HEVC/H.265                                                            | MPEG-4<br>AVC/H.264                                                   |
| Video<br>Encoding | Resolution<br>(DX3224HV)                                              | 4*1080P-50                                                            | 4*1080P-50                                                            | 4*1080P-50                                                            | 4*1080P-25<br>2*1080P-50                                              |
| Video<br>Encoding | Resolution<br>(DX3224HV)                                              | 4*1080P-60/5<br>9.94                                                  | 4*1080P-60,<br>4*1080P-59.9<br>4                                      | 4*1080P-60,<br>4*1080P-59.9<br>4                                      | 4*1080P-30 4<br>*1080P-29.97<br>2*1080P-60 2<br>*1080P-59.94          |

|                |                       |                                                               |                                                               |                                                               |                                                               |
|----------------|-----------------------|---------------------------------------------------------------|---------------------------------------------------------------|---------------------------------------------------------------|---------------------------------------------------------------|
| Video Encoding | Resolution (DX3224HV) | 4*1080I-50                                                    | 4*1080P-50, 4*1080P-25                                        | 4*1080P-50, 4*1080P-25                                        | 4*1080I-50 4*1080P-25 2*1080P-50                              |
| Video Encoding | Resolution (DX3224HV) | 4*1080I-60/59.94                                              | 4*1080P-60, 4*1080P-30                                        | 4*1080P-60, 4*1080P-30                                        | 4*1080I-60 4*1080P-30 2*1080P-60                              |
| Video Encoding | Resolution (DX3224HV) | 4*720P-50                                                     | 4*720P-50                                                     | 4*720P-50                                                     | 4*720P-50                                                     |
| Video Encoding | Resolution (DX3224HV) | 4*720P-60/59.94                                               | 4*720P-60, 4*720P-59.94                                       | 4*720P-60, 4*720P-59.94                                       | 4*720P-60 4*720P-59.94                                        |
| Video Encoding | Resolution (DX3224H)  |                                                               | Output                                                        | Output                                                        | Output                                                        |
| Video Encoding | Resolution (DX3224H)  | Input                                                         | HEVC/H.265                                                    | HEVC/H.265                                                    | MPEG-4 AVC/H.264                                              |
| Video Encoding | Resolution (DX3224H)  | 4*1080P-50                                                    | 4*1080P-50                                                    | 4*1080P-50                                                    | 4*1080P-50                                                    |
| Video Encoding | Resolution (DX3224H)  | 4*1080P-60/59.94                                              | 4*1080P-60, 4*1080P-59.94                                     | 4*1080P-60, 4*1080P-59.94                                     | 4*1080P-60 4*1080P-59.94                                      |
| Video Encoding | Resolution (DX3224H)  | 4*1080I-50                                                    | 4*1080P-50, 4*1080P-25                                        | 4*1080P-50, 4*1080P-25                                        | 4*1080P-25 4*1080P-50 4*1080I-50                              |
| Video Encoding | Resolution (DX3224H)  | 4*1080I-60/59.94                                              | 4*1080P-60, 4*1080P-30                                        | 4*1080P-60, 4*1080P-30                                        | 4*1080P-30 4*1080P-60 4*1080I-60                              |
| Video Encoding | Resolution (DX3224H)  | 4*720P-50                                                     | 4*720P-50                                                     | 4*720P-50                                                     | 4*720P-50                                                     |
| Video Encoding | Resolution (DX3224H)  | 4*720P-60/59.94                                               | 4*720P-60, 4*720P-59.94                                       | 4*720P-60, 4*720P-59.94                                       | 4*720P-60 4*720P-59.94                                        |
| Video Encoding | Chroma                | 4:2:0                                                         | 4:2:0                                                         | 4:2:0                                                         | 4:2:0                                                         |
| Video Encoding | Bitrate (per channel) | 0.5Mbps~20 Mbps (HEVC/H.265) 4 Mbps~20Mbps (MPEG 4 AVC/H.264) | 0.5Mbps~20 Mbps (HEVC/H.265) 4 Mbps~20Mbps (MPEG 4 AVC/H.264) | 0.5Mbps~20 Mbps (HEVC/H.265) 4 Mbps~20Mbps (MPEG 4 AVC/H.264) | 0.5Mbps~20 Mbps (HEVC/H.265) 4 Mbps~20Mbps (MPEG 4 AVC/H.264) |
| Video Encoding | Rate Control          | CBR/VBR                                                       | CBR/VBR                                                       | CBR/VBR                                                       | CBR/VBR                                                       |

|                |                                                                                                                                                        |                                                                                                                                                        |                                                                                                                                                        |                                                                                                                                                        |                                                                                                                                                        |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Video Encoding | GOP Structure                                                                                                                                          | IBBP, IPPP                                                                                                                                             | IBBP, IPPP                                                                                                                                             | IBBP, IPPP                                                                                                                                             | IBBP, IPPP                                                                                                                                             |
| Video Encoding | Advanced Pretreatment                                                                                                                                  | De-interlacing , Noise Reduction, Sharpening                                                                                                           | De-interlacing , Noise Reduction, Sharpening                                                                                                           | De-interlacing , Noise Reduction, Sharpening                                                                                                           | De-interlacing , Noise Reduction, Sharpening                                                                                                           |
| Audio Encoding | Encoding Format                                                                                                                                        | MPEG-1 Layer 2, LC-AAC, HE-AAC, HE-AAC V2, AC3 Passthrough                                                                                             | MPEG-1 Layer 2, LC-AAC, HE-AAC, HE-AAC V2, AC3 Passthrough                                                                                             | MPEG-1 Layer 2, LC-AAC, HE-AAC, HE-AAC V2, AC3 Passthrough                                                                                             | MPEG-1 Layer 2, LC-AAC, HE-AAC, HE-AAC V2, AC3 Passthrough                                                                                             |
| Audio Encoding | Sampling rate                                                                                                                                          | 48KHz                                                                                                                                                  | 48KHz                                                                                                                                                  | 48KHz                                                                                                                                                  | 48KHz                                                                                                                                                  |
| Audio Encoding | Bit-rate (each channel)                                                                                                                                | 48Kbps~384Kbps (MPEG-1 Layer 2 & LC-AAC) 24 Kbps~128 Kbps (HE-AAC) 18 Kbps~56 Kbps (HE-AAC V2)                                                         | 48Kbps~384Kbps (MPEG-1 Layer 2 & LC-AAC) 24 Kbps~128 Kbps (HE-AAC) 18 Kbps~56 Kbps (HE-AAC V2)                                                         | 48Kbps~384Kbps (MPEG-1 Layer 2 & LC-AAC) 24 Kbps~128 Kbps (HE-AAC) 18 Kbps~56 Kbps (HE-AAC V2)                                                         | 48Kbps~384Kbps (MPEG-1 Layer 2 & LC-AAC) 24 Kbps~128 Kbps (HE-AAC) 18 Kbps~56 Kbps (HE-AAC V2)                                                         |
| Audio Encoding | Audio Gain                                                                                                                                             | 0~255                                                                                                                                                  | 0~255                                                                                                                                                  | 0~255                                                                                                                                                  | 0~255                                                                                                                                                  |
| Stream output  | 1 MPTS and 4 SPTS output over UDP/RTP/RTSP per module, Base-T Ethernet interface (1000M/100M self-adaption) (unicast/multicast) IP null packet filterd | 1 MPTS and 4 SPTS output over UDP/RTP/RTSP per module, Base-T Ethernet interface (1000M/100M self-adaption) (unicast/multicast) IP null packet filterd | 1 MPTS and 4 SPTS output over UDP/RTP/RTSP per module, Base-T Ethernet interface (1000M/100M self-adaption) (unicast/multicast) IP null packet filterd | 1 MPTS and 4 SPTS output over UDP/RTP/RTSP per module, Base-T Ethernet interface (1000M/100M self-adaption) (unicast/multicast) IP null packet filterd | 1 MPTS and 4 SPTS output over UDP/RTP/RTSP per module, Base-T Ethernet interface (1000M/100M self-adaption) (unicast/multicast) IP null packet filterd |
| System         | Web based management                                                                                                                                   | Web based management                                                                                                                                   | Web based management                                                                                                                                   | Web based management                                                                                                                                   | Web based management                                                                                                                                   |

|               |                                   |                                   |                                   |                                   |                                   |
|---------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|
| System        | Chinese-English control interface | Chinese-English control interface | Chinese-English control interface | Chinese-English control interface | Chinese-English control interface |
| System        | Ethernet software upgrade         | Ethernet software upgrade         | Ethernet software upgrade         | Ethernet software upgrade         | Ethernet software upgrade         |
| Miscellaneous | Dimension (W× L× H)               | Dimension (W× L× H)               | Dimension (W× L× H)               | 482mm×328mm×44mm                  | 482mm×328mm×44mm                  |
| Miscellaneous | Approx weight                     | Approx weight                     | Approx weight                     | 5kg                               | 5kg                               |
| Miscellaneous | Temperature                       | Temperature                       | Temperature                       | 0~45℃(work), -20~80℃(Storage)     | 0~45℃(work), -20~80℃(Storage)     |
| Miscellaneous | Power                             | Power                             | Power                             | AC 100V-220V±10%, 50/60Hz         | AC 100V-220V±10%, 50/60Hz         |

### Technical Comparison

|                  |                 |                                                                |                                                                     |                                                                     |                                                                                                      |
|------------------|-----------------|----------------------------------------------------------------|---------------------------------------------------------------------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
|                  |                 | FU-4335V-HDMI                                                  | FU-4335V-SDI                                                        | DX3224H                                                             | DX3224HV                                                                                             |
| Input Interface  | Input Interface | 4/8/12*HDMI                                                    | 4/8/12*SDI                                                          | 4/8/12*HDMI/SDI                                                     | 4/8/12*HDMI/SDI                                                                                      |
| Video Encoding   | Video Encoding  | H.265, H.264                                                   | H.265, H.264                                                        | H.265, H.264                                                        | H.265, H.264                                                                                         |
| Audio Encoding   | Audio Encoding  | MPEG-1 Layer 2, LC-AAC, HE-AAC, HE-AAC V2, AC3 Passthrough     | MPEG-1 Layer 2, LC-AAC, HE-AAC, HE-AAC V2, AC3 Passthrough          | MPEG-1 Layer 2, LC-AAC, HE-AAC, HE-AAC V2, AC3 Passthrough          | MPEG-1 Layer 2, LC-AAC, HE-AAC, HE-AAC V2, AC3 Passthrough                                           |
| Input Resolution | H.265           | 3840x2160P @30*2CHs 1920x1080P@50/60*4CHs 1280x720P@50/60*4CHs | 1920x1080P @50/60*4CHs 1280x720P @50/60*4CHs 1920x1080i @50/60*4CHs | 1920x1080P @50/60*4CHs 1920x1080i @50/60*4CHs 1280x720P @50/60*4CHs | 4*1920x1080_60P/59.94P 4*1920x1080_50P 4*1920x1080_60i 4*1920x1080_50i 4*1280x720_60P 4*1280x720_50P |

|                   |                  |                                                               |                                                                     |                                                                     |                                                                                                                                                                                     |
|-------------------|------------------|---------------------------------------------------------------|---------------------------------------------------------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Input Resolution  | H.264            | 3840x2160P @30*1CH 1920x1080P@50/60*2CHs 1280x720P@50/60*4CHs | 1920x1080P @50/60*2CHs 1920x1080i @50/60*2CHs 1280x720P @50/60*4CHs | 1920x1080P @50/60*4CHs 1920x1080i @50/60*4CHs 1280x720P @50/60*4CHs | 4 or 2*1920x1080_60P/59.94P 4 or 2*1920x1080_50P 4 or 2*1920x1080_60i 4 or 2*1920x1080_50i 4*1280x720_60P 4*1280x720_50P                                                            |
| Output Resolution | H.265            | Same as input resolution                                      | 1920x1080P @50/60*4CHs 1280x720P @50/60*4CHs                        | 1920x1080P @50/60*4CHs 1280x720P @50/60*4CHs                        | 4*1920x1080_60P/59.94P 4*1920x1080_50P 4*1920x1080_30P 4*1920x1080_25P 4*1280x720_60P 4*1280x720_50P                                                                                |
| Output Resolution | H.264            | Same as input resolution                                      | 1920x1080P @50/60*2CHs 1280x720P @50/60*4CHs                        | 1920x1080P @50/60*4CHs 1280x720P @50/60*4CHs                        | 2*1920x1080_60P/59.94P (HDMI 3 and HDMI 4) 2*1920x1080_50P (HDMI 3 and HDMI 4) 4*1920x1080_30P/29.97P 4*1920x1080_25P 4*1920x1080_60i 4*1920x1080_50i 4*1280x720_60P 4*1280x720_50P |
| OSD               | OSD              | N/A                                                           | N/A                                                                 | Optional as per order                                               | Optional as per order                                                                                                                                                               |
| Caption Language  | Caption Language | N/A                                                           | N/A                                                                 | Not support caption                                                 | Not support caption                                                                                                                                                                 |

|                                                                                                                      |                                                                                                                      |                                                                                                                      |                                                                                                                      |                                                                                                                      |                                                                                                                      |
|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Note: "CH" means encoding channel, different encoding format can encode different channel with different resolution. | Note: "CH" means encoding channel, different encoding format can encode different channel with different resolution. | Note: "CH" means encoding channel, different encoding format can encode different channel with different resolution. | Note: "CH" means encoding channel, different encoding format can encode different channel with different resolution. | Note: "CH" means encoding channel, different encoding format can encode different channel with different resolution. | Note: "CH" means encoding channel, different encoding format can encode different channel with different resolution. |
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## About Us

FMUSER is a company specializing in broadcasting and television transmission systems and integrating hotel IPTV systems. The company was established in 2009 and has served over 180 countries worldwide. More than 20,000 television stations, radio stations, and hotel clients from different.

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